

## OAC SEAFORTH FIELD BEAN

OAC Seaforth, a backcross-derived anthracnose-resistant cultivar of field bean (*Phaseolus vulgaris* L.) similar in agronomic performance to Seafarer, was developed cooperatively by Agriculture Canada Research Station, Harrow and the Crop Science Department, University of Guelph. OAC Seaforth carries the *Are* gene that confers resistance to alpha, beta, gamma, delta, epsilon and lambda races of anthracnose (*Colletotrichum lindemuthianum* (Sacc. and Magn.) Briosi and Cav.)

Key words: Cultivar description, anthracnose resistance, bean (white)

[Haricots OAC Seaforth.]

Titre abrégé: Haricots OAC Seaforth.

OAC Seaforth, cultivar de haricot (*Phaseolus vulgaris* L.) résistant à l'anthracnose, obtenu par rétrocroisement et possédant des qualités agronomiques analogues à celles de Seafarer, a été créé conjointement par la station fédérale de recherches agricoles de Harrow et le département des sciences agricoles de l'Université de Guelph. OAC Seaforth porte le gène *Are* qui commande la résistance aux souches alpha, beta, gamma, delta, epsilon et lambda de l'anthracnose (*Colletotrichum lindemuthianum* (Sacc. et Magn.) Briosi et Cav.).

Mots clés: Description de cultivar, résistance à l'anthracnose, haricot (blanc)

Bean anthracnose, caused by *Colletotrichum lindemuthianum* (Sacc. and Magn.) Briosi and Cav., has not caused losses to white bean producers in North America for the past two decades because commonly grown cultivars were resistant to the previously prevalent races (alpha, beta and gamma). An outbreak of this disease in southern Ontario during 1976 was caused largely by the delta race and to a lesser extent by the lambda race, both new to Canada. A backcross breeding program was initiated in 1977 to improve the anthracnose resistance of Seafarer.

License no. 2327 was issued for OAC Seaforth on 2 May 1983 by the Plant Health and Plant Directorate, Agriculture Canada. This paper describes the origin, characteristics and performance of this cultivar.

### Pedigree Breeding

OAC Seaforth originated from a cross and subsequent backcrosses of Seafarer (recurrent parent) × PI 326,418 (donor parent used as the source of *Are* gene for resistance to several races of anthracnose). OAC Seaforth represents homozygous (*Are/Are*)

progeny of the sixth backcross (BC<sub>6</sub>). Individual backcross families were purified and bulked as Seafarer ARE (also designated as Seafarer 80 or 83 in some cases).

### Agronomic Performance

OAC Seaforth has been tested in Ontario Field Bean Variety Trials since 1981 (Table 1). Cooperative trials were conducted by the Colleges of Agricultural Technology at Ridgetown and Centralia; Crop Science Dep., University of Guelph and Research Station, Agriculture Canada, Harrow. Over the 2-yr period 1981–1982, OAC Seaforth averaged 2 days later in maturity than Seafarer and yielded 5% less than Seafarer. It is adapted as an early-maturing cultivar to the main field bean growing areas in southern Ontario with 2500–2700 corn heat units.

### Description

OAC Seaforth is a bush (determinate) plant type with characteristics very similar to Seafarer. OAC Seaforth has white flowers, cream tan pods, and white seeds similar in size and shape to Seafarer.

### Quality

Taste panel evaluation of canned beans

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Table 1. Average performance of OAC Seaforth and Seafarer in 12 Ontario Field Bean Variety Trials grown at Brussels, Centralia, Dresden, Elora, Kenwood, Kippen and Ridgeway, 1981–1982

| Cultivar     | Yield<br>(tonnes/ha) | Maturity<br>(days) | Seed<br>weight<br>(g/100 seeds) | Cooking<br>quality<br>score† | White mold<br>(% of plants<br>infected)‡ |
|--------------|----------------------|--------------------|---------------------------------|------------------------------|------------------------------------------|
| OAC Seaforth | 2.27                 | 90                 | 20.1                            | 2.5                          | 33                                       |
| Seafarer     | 2.38                 | 88                 | 19.3                            | 2.2                          | 35                                       |

†Values shown are combined quality indices including factors for appearance, flavor, and texture of canned beans, based on samples from six trials (1981–1982) and rated on a scale from 1 (poor) to 5 (excellent). Tests were conducted at the Agriculture Canada Research Station, Harrow, Ontario.

‡Mean of four trials infected with white mold: Dresden, Kerwood and Ridgeway in 1981; Mitchell in 1982.

Table 2. Disease ratings of OAC Seaforth and Seafarer for races of anthracnose and Bean Common Mosaic Virus (BCMV)†

| Cultivar     | Anthracnose              |      |       |         |       |        | BCMV |    |
|--------------|--------------------------|------|-------|---------|-------|--------|------|----|
|              | alpha                    | beta | gamma | epsilon | delta | lambda | 1    | 15 |
|              | <i>Disease severity‡</i> |      |       |         |       |        |      |    |
| OAC Seaforth | 0                        | 0    | 0     | 0       | 0     | 0      | R§   | R  |
| Seafarer     | 0                        | 0    | 6     | 0       | 7     | 9      | R    | R  |

†Based on controlled environment test conducted at the Agriculture Canada Research Station, Harrow, Ontario in 1981–1982.

‡0 = resistant, 9 = highly susceptible.

§R = resistant, S = susceptible.

conducted by the Agriculture Canada Research Station, Harrow, Ontario (Table 1) and by Libby, McNeil, and Libby Ltd., Chatham, Ontario, rated OAC Seaforth as acceptable and comparable with Seafarer in processed (canned) products. Protein content was also similar to Seafarer and typical of white beans, navy beans or Michigan pea beans. Seed size and shape are acceptable to the industry.

### Disease Reaction

OAC Seaforth is resistant to races 1 and 15 of bean common mosaic virus, and resistant to alpha, beta, delta, epsilon, gamma, and lambda races of anthracnose (Table 2). OAC Seaforth is similar to Seafarer in susceptibility to white mold caused by *Sclerotinia sclerotiorum* (Lib.) De Bary (Table 1).

### Pedigree Seed Stocks

Breeder seed was released through the

Ontario Seed Stock Distribution Committee of the Canadian Seed Growers' Association. The crop Science Department, University of Guelph will maintain breeder seed.

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